



## 9547 - NIRCam and MIRI observations of a barred galaxy

Cycle: 4, Proposal Category: DD

### INVESTIGATORS

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### OBSERVATIONS

| <i>Folder</i>      | <i>Observation</i> | <i>Label</i>        | <i>Observing Template</i> | <i>Science Target</i> |
|--------------------|--------------------|---------------------|---------------------------|-----------------------|
| Observation Folder |                    |                     |                           |                       |
|                    | 1                  | BarredGalaxy-NIRCam | NIRCam Imaging            | (1) Barred-Galaxy     |
|                    | 2                  | BarredGalaxy-MIRI   | MIRI Imaging              | (1) Barred-Galaxy     |

### ABSTRACT

We will observe this barred galaxy on 4 NIRCam narrow and medium filters, and 3 MIRI filters. Observations are sufficiently deep to allow for the creation of low-noise color-composite images. The target is symmetric enough that no PA restriction is needed; it only needs one NIRCam module, and a 1x2 MIRI mosaic

### OBSERVING DESCRIPTION

## JWST Proposal 9547 (Created: Wednesday, November 5, 2025, 6:00:09PM Eastern Standard Time) - Overview

We will observe this barred galaxy on 4 NIRCам narrow and medium filters, and 3 MIRI filters. Observations are sufficiently deep to allow for the creation of low-noise color-composite images. The target is symmetric enough that no PA restriction is needed; it only needs one NIRCам module, and a 1x2 MIRI mosaic

# Proposal 9547 - Targets - NIRCам and MIRI observations of a barred galaxy

| Fixed Targets | #                                                                                                                                                            | Name          | Target Coordinates                                                                 | Targ. Coord. Corrections | Miscellaneous |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------|--------------------------|---------------|
|               | (1)                                                                                                                                                          | Barred-Galaxy | RA: 13 52 56.7998 (208.2366658d)<br>Dec: +02 48 46.04 (2.81279d)<br>Equinox: J2000 | Epoch of Position: 2000  |               |
|               | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=Galaxy<br>Description=[Barred spiral galaxies] |               |                                                                                    |                          |               |

# Proposal 9547 - Observation 1 - NIRCам and MIRI observations of a barred galaxy

|                      |                                                                                                                                                                                                                                                                                                                  |                     |                                                                                    |                 |            |                          |                    |               |                     |                    |                              |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------|-----------------|------------|--------------------------|--------------------|---------------|---------------------|--------------------|------------------------------|
| Observation          | Proposal 9547, Observation 1: BarredGalaxy-NIRCam                                                                                                                                                                                                                                                                |                     |                                                                                    |                 |            |                          |                    |               |                     |                    | Wed Nov 05 23:00:09 GMT 2025 |
|                      | Diagnostic Status: Warning                                                                                                                                                                                                                                                                                       |                     |                                                                                    |                 |            |                          |                    |               |                     |                    |                              |
|                      | Observing Template: NIRCam Imaging                                                                                                                                                                                                                                                                               |                     |                                                                                    |                 |            |                          |                    |               |                     |                    |                              |
| Diagnostics          | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                                                                      |                     |                                                                                    |                 |            |                          |                    |               |                     |                    |                              |
| Fixed Targets        | #                                                                                                                                                                                                                                                                                                                | Name                | Target Coordinates                                                                 |                 |            | Targ. Coord. Corrections |                    |               | Miscellaneous       |                    |                              |
|                      | (1)                                                                                                                                                                                                                                                                                                              | Barred-Galaxy       | RA: 13 52 56.7998 (208.2366658d)<br>Dec: +02 48 46.04 (2.81279d)<br>Equinox: J2000 |                 |            | Epoch of Position: 2000  |                    |               |                     |                    |                              |
|                      | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.                                                                                                                                                                                                                |                     |                                                                                    |                 |            |                          |                    |               |                     |                    |                              |
|                      | Category=Galaxy<br>Description=[Barred spiral galaxies]                                                                                                                                                                                                                                                          |                     |                                                                                    |                 |            |                          |                    |               |                     |                    |                              |
| Template             | Module                                                                                                                                                                                                                                                                                                           |                     |                                                                                    |                 |            | Subarray                 |                    |               |                     |                    |                              |
|                      | B                                                                                                                                                                                                                                                                                                                |                     |                                                                                    |                 |            | FULL                     |                    |               |                     |                    |                              |
| Dithers              | #                                                                                                                                                                                                                                                                                                                | Primary Dither Type |                                                                                    | Primary Dithers |            | Subpixel Dither Type     |                    | Dither Size   |                     | Subpixel Positions |                              |
|                      | 1                                                                                                                                                                                                                                                                                                                | INTRAMODULEBOX      |                                                                                    | 6               |            | STANDARD                 |                    |               |                     | 1                  |                              |
| Spectral Elements    | #                                                                                                                                                                                                                                                                                                                | Short Filter        | Long Filter                                                                        | Readout Pattern | Groups/Int | Integrations/Exp         | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID    |                              |
|                      | 1                                                                                                                                                                                                                                                                                                                | F182M               | F335M                                                                              | SHALLOW4        | 7          | 1                        | 6                  | 6             | 2190.301            |                    |                              |
|                      | 2                                                                                                                                                                                                                                                                                                                | F090W               | F277W                                                                              | BRIGHT2         | 6          | 1                        | 6                  | 6             | 773.047             |                    |                              |
| Special Requirements | Aperture PA Range 0.05262691 to 195.05262691 Degrees (V3 0.0 to 195.0)<br>Aperture PA Range 206.05262691 to 217.05262691 Degrees (V3 206.0 to 217.0)<br>Aperture PA Range 224.05262691 to 282.05262691 Degrees (V3 224.0 to 282.0)<br>Aperture PA Range 288.05262691 to 359.05262691 Degrees (V3 288.0 to 359.0) |                     |                                                                                    |                 |            |                          |                    |               |                     |                    |                              |

# Proposal 9547 - Observation 2 - NIRCam and MIRI observations of a barred galaxy

|                                  |                                                                                                   |               |                                  |                  |                  |                |                          |               |                    |                     |                              |
|----------------------------------|---------------------------------------------------------------------------------------------------|---------------|----------------------------------|------------------|------------------|----------------|--------------------------|---------------|--------------------|---------------------|------------------------------|
| Observation                      | Proposal 9547, Observation 2: BarredGalaxy-MIRI                                                   |               |                                  |                  |                  |                |                          |               |                    |                     | Wed Nov 05 23:00:09 GMT 2025 |
|                                  | Diagnostic Status: Warning                                                                        |               |                                  |                  |                  |                |                          |               |                    |                     |                              |
| Observing Template: MIRI Imaging |                                                                                                   |               |                                  |                  |                  |                |                          |               |                    |                     |                              |
| Diagnostics                      | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.       |               |                                  |                  |                  |                |                          |               |                    |                     |                              |
|                                  | (Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.       |               |                                  |                  |                  |                |                          |               |                    |                     |                              |
| Fixed Targets                    | #                                                                                                 | Name          | Target Coordinates               |                  |                  |                | Targ. Coord. Corrections |               |                    | Miscellaneous       |                              |
|                                  | (1)                                                                                               | Barred-Galaxy | RA: 13 52 56.7998 (208.2366658d) |                  |                  |                | Epoch of Position: 2000  |               |                    |                     |                              |
|                                  |                                                                                                   |               | Dec: +02 48 46.04 (2.81279d)     |                  |                  |                |                          |               |                    |                     |                              |
|                                  |                                                                                                   |               | Equinox: J2000                   |                  |                  |                |                          |               |                    |                     |                              |
|                                  | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. |               |                                  |                  |                  |                |                          |               |                    |                     |                              |
|                                  | Category=Galaxy                                                                                   |               |                                  |                  |                  |                |                          |               |                    |                     |                              |
|                                  | Description=[Barred spiral galaxies]                                                              |               |                                  |                  |                  |                |                          |               |                    |                     |                              |
| Template                         | Subarray                                                                                          |               |                                  |                  |                  |                |                          |               |                    |                     |                              |
|                                  | FULL                                                                                              |               |                                  |                  |                  |                |                          |               |                    |                     |                              |
| Mosaic                           | Rows                                                                                              | Columns       | Row Overlap %                    |                  | Column Overlap % |                | Row shift (deg)          |               | Column shift (deg) |                     | Tile Order                   |
|                                  | 1                                                                                                 | 2             | 10.0                             |                  | 10.0             |                | 0.0                      |               | 0.0                |                     | DEFAULT                      |
| Dithers                          | #                                                                                                 | Dither Type   | Starting Point                   | Number of Points | Points           | Starting Set   | Number of Sets           | Optimized For | Direction          | Pattern Size        |                              |
|                                  | 1                                                                                                 | CYCLING       | 4                                | 6                |                  |                |                          |               |                    | DEFAULT             |                              |
| Spectral Elements                | #                                                                                                 | Filter        | Readout Pattern                  | Groups/Int       | Integrations/Exp | Exposures/Dith | Dither                   | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID              |
|                                  | 1                                                                                                 | F770W         | FASTR1                           | 50               | 3                | 1              | Dither 1                 | 6             | 18                 | 2530.836            |                              |
|                                  | 2                                                                                                 | F1000W        | FASTR1                           | 50               | 3                | 1              | Dither 1                 | 6             | 18                 | 2530.836            |                              |
|                                  | 3                                                                                                 | F1280W        | FASTR1                           | 50               | 3                | 1              | Dither 1                 | 6             | 18                 | 2530.836            |                              |
| Special Requirements             | Group Visits within 53.0 Days                                                                     |               |                                  |                  |                  |                |                          |               |                    |                     |                              |
|                                  | Visits Same PA                                                                                    |               |                                  |                  |                  |                |                          |               |                    |                     |                              |